

Arithmetic balancing using logarithmically many fair-block profiles

Argument by the rational-design research agent; written and checked independently

30 September 2026

Scope. This note concerns only the arithmetic of block multiplicities. It does not prove that concatenating the resulting fair blocks preserves permutation fairness, and therefore does not resolve polynomial equalization. Indeed, n cyclic relabelings of any word already balance its multiplicities, with total length n times the original length. The point here is to reduce the number of block profiles to $O(\log n)$ while retaining polynomial total length.

Lemma 1. *Let $c_1, \dots, c_n$ be positive integers and $q \geq 1$. There are permutations $\sigma_1, \dots, \sigma_q$ for which*

$$L := \text{lcm}_{i \in [n]} \gcd(c_{\sigma_1(i)}, \dots, c_{\sigma_q(i)}) \leq \left(\prod_{i=1}^n c_i \right)^{n^{1/q}/n}.$$

Proof. Choose the q permutations independently and uniformly. For a prime p and $a \geq 1$, let $k_{p,a}$ count the indices j with $p^a \mid c_j$, and put $x = k_{p,a}/n$. The event $p^a \mid L$ requires at least one coordinate i whose q selected multiplicities are all divisible by p^a . The union bound gives

$$\Pr(p^a \mid L) \leq \min(1, nx^q) \leq n^{1/q}x.$$

The final inequality follows by splitting at $x = n^{-1/q}$. Therefore

$$\mathbb{E} \log L \leq \frac{n^{1/q}}{n} \sum_{p,a} k_{p,a} \log p = \frac{n^{1/q}}{n} \sum_i \log c_i.$$

At least one tuple of permutations has logarithm no larger than its expectation, proving the result. $\square$

Lemma 2 (A positive coin bound). *Let $a_1, \dots, a_q \leq H$ be positive integers of greatest common divisor g . Every multiple C of g with $C \geq qH + H^2$ has a representation $C = \sum_{t=1}^q b_t a_t$ with all b_t positive integers.*

Proof. Subtract one copy of every generator. It remains to represent a multiple C' of g with $C' \geq H^2$ by nonnegative coefficients. Divide the generators by g , and let a be their minimum. The directed residue graph modulo a , with each generator allowed as a step, is strongly connected because the scaled gcd is one. A shortest path from zero to any residue has at most $a - 1$ edges. It therefore represents that residue by a nonnegative combination of scaled generators of total value at most $(a - 1)H/g$. In the original units this value is at most $(a - 1)H \leq H^2$. Choose the residue of C'/g and add copies of the minimum generator to reach C' . Restoring the initially subtracted generators makes all coefficients positive. $\square$

Theorem 1. *Let s be a permutation-fair word on $n \geq 2$ letters, with multiplicities c_i , length $A = \sum_i c_i$, and $H = \max_i c_i$. There are $q = \lceil \log_2 n \rceil$ permutation-fair words $s_1, \dots, s_q$, each obtained by relabeling s and independently blowing up occurrences of each letter, whose combined multiplicity of every letter is the same integer C , with*

$$C \leq qH + 2H^2, \quad \sum_t |s_t| = nC \leq 3A^3.$$

Proof. Use Lemma 1. Since $n^{1/q} \leq 2$ and the geometric mean is at most H , its resulting L satisfies $L \leq H^2$. Let C be the least multiple of L no smaller than $qH + H^2$. Then $C \leq qH + 2H^2$. For each coordinate i , its q generators $c_{\sigma_t(i)}$ have gcd dividing L , hence dividing C . Lemma 2 gives positive integers b_{ti} with

$$\sum_{t=1}^q b_{ti} c_{\sigma_t(i)} = C.$$

Relabel s according to σ_t , then replace every occurrence of letter i by b_{ti} consecutive copies. Every full-pattern count is multiplied by $\prod_i b_{ti}$, so each resulting word s_t is still permutation-fair. Its i -multiplicity is $b_{ti} c_{\sigma_t(i)}$, giving the asserted common total. Finally $n, q, H \leq A$, so $n(qH + 2H^2) \leq 3A^3$. $\square$

The remaining geometric obstruction. The concatenation $s_1 \cdots s_q$ is balanced in multiplicities, but its order probabilities include selections spanning different blocks. Their contribution is not controlled by the arithmetic identities above. A construction that eliminated these cross-block defects at cost exponential only in the number q of profiles would make the lemma relevant to polynomial equalization. No such construction is established here. There is an additional concrete barrier: if only these q profile directions are

reordered or repeated, the block-profile matrix has rank at most q . The independently proved continuous CDF-rank theorem in `../size_bounds/two_dimensional_equalization_barrier.tex` then forces at most q distinct normalized letter profiles, with at most one profile allowed to repeat. Thus at least $n - q + 1$ letters must share one profile whenever such a concatenation is three-wise fair. For generic output profiles this fails, regardless of the number of repetitions. Any eventual geometric use of the arithmetic lemma would therefore need new profile directions or additional shared structure.